3: Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
∴ The condition that for each column having a leading entry, all positions below the leading entry must contain 0 is True.

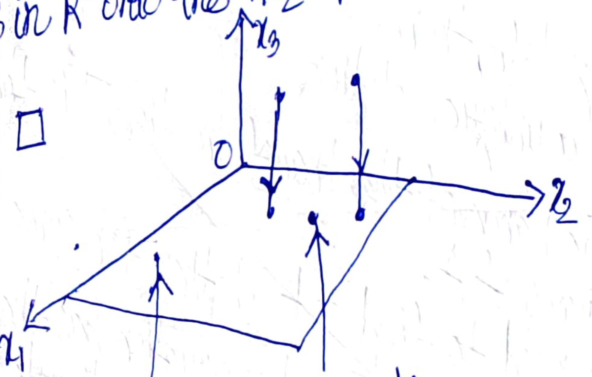
∴ The matrix $\begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 5/2 \end{bmatrix}$ is in echelon form.

There is a row ($[0\ 0\ 5/2]$) of the matrix in the form $[0\ 0\ 0 \ldots\ldots 0\ b]$ where $b \neq 0$. By Theorem 2, the system of equations is inconsistent, i.e. no solution set exists.

$\boxed{\therefore c \notin T(x)}$ □

Example 2: If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then the transformation $x \mapsto Ax$ projects points in $\mathbb{R}^3$ onto the x_1x_2-plane because

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$ □



A projection transformation

Example 3: Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. The transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x) = Ax$ is called a $\boxed{\text{shear transformation}}$. It can be shown that if T acts on each point in the 2×2 square shown in Figure 4, then the set of images forms the shaded parallelogram.

- The key idea is to show that T maps line segments onto line segments (as shown in Exercise 27) and then to the check that the corners of the square map onto the vertices of the parallelogram.
- For instance, the image of the point $u = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ is $T(u) = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$, and the image of $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ is $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$.
- T deforms the square as if the top of the square were pushed to the right while the base is held fixed. □